Predicting 48-Hour Extubation Failure Using Pre-Extubation Clinical Data: A Retrospective MIMIC-IV Study

Jiajian He (1357602), Wenhao Duan (1695089), Haoran Zhang (1697370), Kun Xing (1773853), Zongcheng Jiang (1653403)

Group 25

ABSTRACT

Objective: To predict 48-hour extubation failure from pre-extubation data and compare multivariable models with the rapid shallow breathing index (RSBI).

Materials and Methods: Adults undergoing their first planned ICU extubation in MIMIC-IV v3.1 (n = 17,146) were studied retrospectively. Predictors comprised six-hour pre-extubation physiological means and static clinical variables. Failure was recorded reintubation or in-hospital death within 48 hours during the same admission. Logistic regression and XGBoost were developed using earlier-period data and evaluated in a later temporal test cohort for discrimination, calibration and potential clinical net benefit.

Results: In 897 test patients (106 failures), LR and XGBoost both had an area under the receiver operating characteristic curve (AUROC) of 0.835, and the paired bootstrap 95% confidence interval for their difference included zero. At the threshold selected using development data, LR had 90.6% sensitivity and 97.9% negative predictive value. Both showed higher observed discrimination than RSBI in 707 patients with available measurements. Missingness-related exclusions accounted for 35.0% of pre-filter test failures.

Discussion: XGBoost showed no clear advantage over LR. Discrimination was higher for death than for reintubation. Missingness-related selection limits applicability, and predictive performance does not establish benefit from clinical implementation.

Conclusion: Pre-extubation data supported retrospective risk prediction. External and prospective validation are needed before clinical use.

Key words: Extubation failure; Mechanical ventilation; Machine learning; Logistic regression; MIMIC-IV

1 INTRODUCTION

Discontinuing invasive ventilation requires balancing premature extubation against prolonged ventilatory support. A multicentre study reported hospital mortality of 42.3% among patients reintubated within 48 hours, compared with 14.0% among those without reintubation[1]. The intended use is to provide ICU clinicians with risk estimates at planned extubation, alongside routine assessment, to identify patients who may need closer post-extubation monitoring and reassessment. This use requires prospective evaluation.

Current assessment includes spontaneous breathing trials and respiratory and haemodynamic evaluation[2]. However, individual indices provide incomplete information. A meta-analysis found that a rapid shallow breathing index (RSBI) below 105 predicted successful extubation with pooled sensitivity of 0.83 and specificity of 0.58[3]. Multivariable machine-learning approaches show promise[4,5], but heterogeneous outcomes and limited external and prospective validation complicate interpretation[6]. Evaluation should consider calibration and potential clinical benefit alongside discrimination.

This retrospective MIMIC-IV v3.1 study included adults at their first planned ICU extubation following invasive ventilation. Six-hour physiological means increased measurement availability while retaining recent information, consistent with previous research[7]. Regularised logistic regression (LR) provided an interpretable reference, while XGBoost captured nonlinear relationships and interactions. Failure comprised recorded reintubation or in-hospital death within 48 hours during the same admission; non-invasive ventilation alone was excluded.

We compared multivariable prediction with six-hour mean RSBI in a temporal test cohort, assessing discrimination, calibration and potential clinical net benefit. Sensitivity analysis examined retaining patients excluded by the missingness filter.

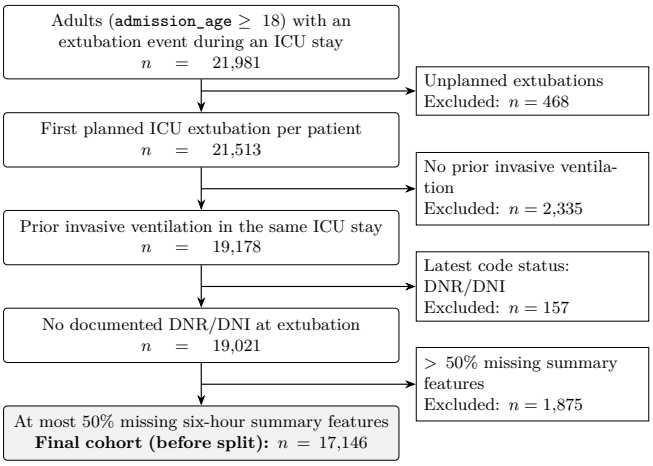


Figure 1. Cohort selection for the six-hour average dataset. Only the first planned extubation per patient was retained. Patients with undocumented code status were retained. Final cohort split into 15,493 development patients and 897 test patients, excluding 756 whose intervals spanned 2020. DNR/DNI: do-not-resuscitate or do-not-intubate.

2 METHODS

2.1 Dataset

Data were extracted from MIMIC-IV v3.1 using Google BigQuery [8,9]. Adults aged ≥ 18 years were included at their first planned ICU extubation, requiring prior invasive ventilation in the same ICU stay. Unplanned extubations and patients with documented do-not-resuscitate or do-not-intubate (DNR/DNI) status were excluded; undocumented status was retained. Treatment limitations may influence reintubation decisions and outcome interpretation. Figure 1 summarises cohort selection, and Table A.1 specifies source tables, codes and temporal rules.

Following previous MIMIC-IV research [7], extubation failure comprised recorded reintubation or in-hospital death within 48 hours after extubation during the same admission; non-invasive ventilation alone was excluded. Procedure-based identification may miss reintubations, and post-discharge events were not captured, limiting ascertainment of event-free follow-up.

2.2 Data Preprocessing

Extubation time t anchored extraction of vital signs and ventilator measurements from `chartevents`, and confirmed arterial blood-gas results from `labevents`, within $[t-6\,\mathrm{h}, t)$. Measurements required `storetime` $\leq t$ to restrict predictors to information available by extubation. The outcome window was $(t, t+48\,\mathrm{h}]$. Variable-specific validity checks and unit conversions were applied; extraction rules, units and item IDs are provided in Tables A.2 and A.3.

Six-hour pre-extubation averages of vital signs, arterial blood-gas measurements and ventilator parameters have been used in previous MIMIC-IV research [7]. Each physiological feature was averaged across available valid observations, producing one row per patient without hourly binning or forward filling. Features without observations remained missing. Respiratory indices were calculated before averaging:

$$\mathrm{RSBI} = \frac{\mathrm{RR}_{\mathrm{spont}}}{V_{T,\mathrm{spont}}/1000}, \qquad \mathrm{P/F} = \frac{\mathrm{PaO_2}}{\mathrm{FiO_2}}. \tag{1}$$

RSBI paired spontaneous respiratory rate (breaths/min) and tidal volume (mL) at the same timestamp; P/F paired $\mathrm{PaO_2}$ and fractional $\mathrm{FiO_2}$ from the same arterial specimen. Static predictors comprised age, sex, ICU admission type, and elapsed hospital and ICU stay before extubation.

Patients missing more than 50% of the 19 physiological summaries were excluded before imputation, retaining 17,146 of 19,021 eligible patients (90.1%). This assessed feature availability rather than continuous measurement coverage. Missingness varied substantially, reaching 99.8% for bicarbonate; feature-level percentages are reported in Table A.2, and Table A.4 compares retained and excluded patients.

2.3 Models

LR serves as an interpretable model with L2 regularization and linear effects on log odds. XGBoost uses an ensemble of boosted decision trees to capture nonlinear relationships and interactions [10].

Random guessing and six-hour mean RSBI provided chance and clinical baselines respectively [3]. Untuned models used fixed hyperparameters while keeping preprocessing, weighting, calibration and folds constant. These comparisons were added after inspecting the results from the tuned models and were exploratory.

The development and test split used calendar-year intervals at the time of extubation, derived from MIMIC-IV shifted dates and anchor-year information [9]. Intervals ending before 2020 were used for development, whereas intervals starting in 2020 or later were used for testing. Intervals spanning the cutoff were excluded, with no overlap between patients.

Nested cross-validation separated hyperparameter tuning from development-stage performance estimation [11]. We used five outer folds and three inner folds with shuffled stratification. Inner-fold AUROC selected hyperparameters, while mean outer-fold AUROC selected between tuned LR and XGBoost. Missing numerical values are filled with the training fold median and zero used for entirely missing training columns. Missing categorical values were filled with 'unknown' before one-hot encoding and numerical predictors were standardised for LR. Class weights were calculated for each

Table 1. Model performance in all 897 filtered test patients.

A. Discrimination, probability prediction error and calibration

Model	AUROC (95% CI)	AP (95% CI)	Brier score (95% CI)	Scaled Brier score (95% CI)	O:E ratio	Calibration slope
LR	0.835 (0.794 to 0.873)	0.469 (0.379 to 0.573)	0.0805 (0.0682 to 0.0931)	0.227 (0.145 to 0.301)	0.92	1.01
XGBoost	0.835 (0.795 to 0.874)	0.507 (0.414 to 0.606)	0.0805 (0.0689 to 0.0925)	0.228 (0.160 to 0.289)	0.91	1.17
Random guessing	0.501 (0.444 to 0.555)	0.115 (0.093 to 0.145)	—	—	—	—

Notes: The event-rate reference for AP is 0.118. A lower Brier score and a higher scaled Brier score are better; the ideal value for the O:E ratio and calibration slope is 1. O:E and calibration slope are point estimates without confidence intervals. AP: average precision; AUROC: area under the receiver operating characteristic curve; CI: confidence interval; LR: logistic regression; NPV: negative predictive value; O:E: observed-to-expected; PPV: positive predictive value.

B. Performance of the two models at their frozen thresholds

Model	Frozen threshold	Sensitivity, % (95% CI)	Specificity, % (95% CI)	PPV, % (95% CI)	NPV, % (95% CI)
LR	0.08261	90.6 (84.4 to 95.7)	59.8 (56.1 to 63.2)	23.2 (18.9 to 27.3)	97.9 (96.5 to 99.0)
XGBoost	0.08322	86.8 (79.8 to 92.9)	57.1 (53.6 to 60.6)	21.3 (17.2 to 25.2)	97.0 (95.3 to 98.5)

training split as shown in Table B.1. As weighting might influence predicted probabilities, sigmoid calibration is done using cross-validated predictions generated within the training partition [12].

Standardised numerical LR coefficients and Tree-SHAP for tuned XGBoost are calculated using development patients. These show model associations and contributions to predictions and not causal effects [13].

After inspecting the primary test result, we added an exploratory sensitivity analysis without the missingness filter. The primary analysis is unchanged. We then compared models trained on the full and filtered cohort using the same 897 retained test patients. Models trained on the full cohort were also evaluated with the 196 excluded test patients.

2.4 Evaluation Metrics

897 filtered test patients were evaluated using AUROC and average precision (AP). AP is used for imbalanced outcomes and comparison with outcome prevalence [14]. Thresholds are selected from development out-of-fold predictions to maximise specificity with at least 80% sensitivity, then fixed for testing. This exploratory aim limits false alerts, and prioritises failure detection due to the potential consequence of missed failure. Sensitivity, specificity and positive and negative predictive values are calculated.

Brier and scaled Brier scores assess probability accuracy. Calibration curves, slopes and observed-to-expected ratios compare predictions with observed outcomes [15]. We use 1000 patient bootstrap resamples to calculate 95% confidence intervals for discrimination, classification and Brier scores. Paired bootstrap differences in AUROC, AP and Brier scores compare tuned LR with tuned XGBoost, and tuned with untuned versions of each model.

Decision-curve analysis assessed potential clinical net benefit at risk thresholds of 1-30% [16]. All 4 models, RSBI ≥ 105, act on all and act on none are compared using the same 707 patients with RSBI records. The established RSBI cutoff is applied to six-hour mean RSBI [3].

2.5 Ethics Statement

This study used de-identified MIMIC-IV v3.1 data accessed under PhysioNet's credentialing and data-use requirements [8]. No patients were contacted and no attempt was made to re-identify individuals. Patient-level source or derived data will not be redistributed; only aggregate results and analysis code will be shared.

3 RESULTS

3.1 Model Performance

Of the 897 patients in the main test set, 106 had extubation failure. Tuned XGBoost was selected during the development cross-validation with mean outer-fold AUROC 0.827 (SD 0.011), as compared with 0.812 (SD 0.013) for tuned LR. The tuned XGBoost and LR both had an AUROC of about 0.835 (Table 1; Figure 2). Their AP was 0.507 and 0.469, and their Brier score was about 0.0805 for both. Both had an O:E slightly

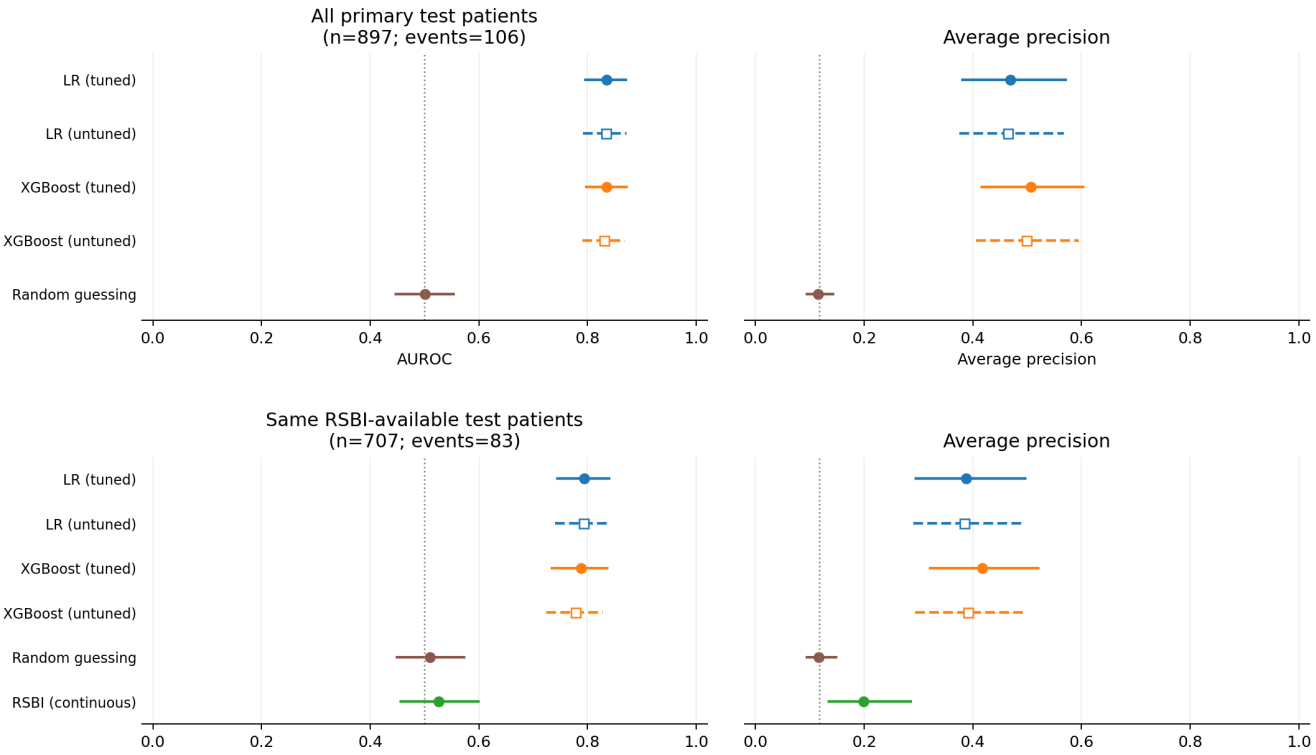


Figure 2. Discrimination comparison in the temporal test set. The top row shows all 897 Filtered test patients (106 events), and the bottom row shows the 707 of these patients with RSBI records (83 events). Horizontal bars show patient bootstrap 95% CIs. RSBI uses its continuous value, and the AP reference line is the event rate in each population. AP: average precision; AUROC: area under the receiver operating characteristic curve; CI: confidence interval; RSBI: rapid shallow breathing index.

below 1, which shows that the overall predicted risk was slightly higher than the observed risk. At the frozen threshold, XGBoost missed 14 cases and gave 339 false alerts; LR missed 10 cases and gave 318 false alerts, with a sensitivity of 86.8% and 90.6%, respectively.

For the tuned XGBoost and LR, the intervals for the differences in AUROC, AP and Brier score all included 0, showing no clear advantage of one algorithm over the other. In the tuned-vs-untuned comparisons, AUROC differences were +0.001 for LR and +0.005 for XGBoost. The AP differences also had intervals including zero. Thus, there is no clear improvement in discrimination after tuning.

In the same 707 patients with RSBI records, the tuned XGB and LR were better than continuous RSBI at telling which patients would have extubation failure (Figure 2). DCA showed that, at some risk thresholds, using model alerts had more potential benefit than using the RSBI ≥ 105 rule or acting on all patients (Figure 3). ICU duration before extubation and ICU type at admission had the largest mean absolute SHAP contributions in XGBoost, followed by age, PEEP and respiratory rate (Figure 4B). In LR, spontaneous tidal volume had the largest absolute numerical coefficient, which was negative (Figure 4A).

3.2 Error Analysis

The 791 patients with neither outcome were used as common controls. AUROC for death was higher than for reintubation: 0.912 versus 0.741 for tuned XGB, and 0.885 versus 0.766 for LR. The case groups differed between the two outcomes; these are descriptive comparisons without a formal test of the difference.

3.3 Sensitivity Analysis

In the same 897 retained test patients, the AUROC difference for Full-trained minus Filtered-trained models was +0.003 for tuned LR (95% CI: −0.003 to +0.008) and −0.007 for XGBoost (−0.016 to +0.003). Five of the six paired differences in AUROC, AP and Brier score had intervals including 0. LR's AP increased by 0.017 (+0.005 to +0.030), but there was no consistent benefit across algorithms and measures. Among the 196 excluded test patients, 57 had failure (29.1%); the AUROC of Full-trained LR and XGBoost was 0.710 and 0.723, respectively.

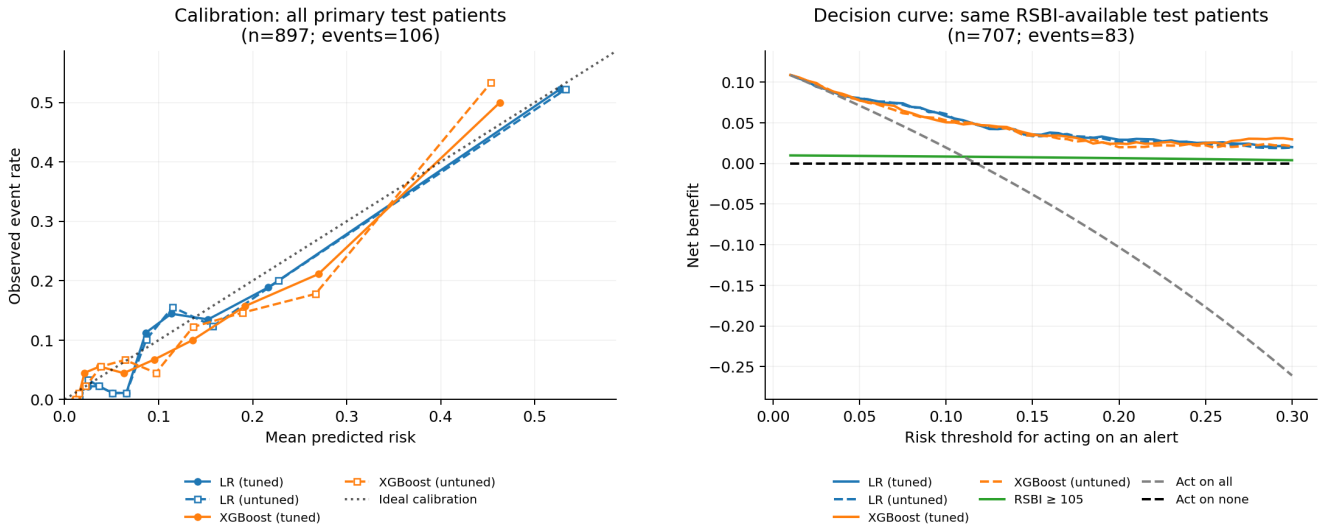


Figure 3. Primary analysis. Calibration in all 897 Filtered test patients and DCA in the 707 patients with RSBI records. The left panel shows calibration curves for all 897 patients (106 events), divided into 10 groups by each model's predicted risk. The right panel shows DCA for the 707 patients with RSBI records (83 events), comparing all four models, RSBI $\geq$ 105, act on all and act on none. No confidence intervals were calculated for net benefit. The patient counts are shown separately for the two panels; the DCA results for 707 patients must not be treated as results for all 897 patients.

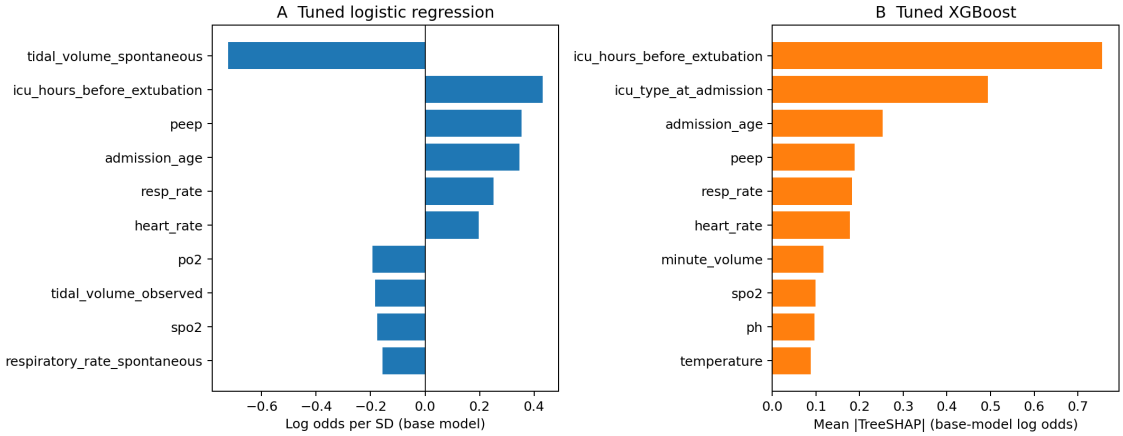


Figure 4. (A) 10 numerical LR coefficients ranked by absolute magnitude (log odds per SD). (B) 10 top XGBoost variables by mean absolute TreeSHAP value (log odds). Both use development data and pre-calibration models. Categorical SHAP values are aggregated by source variable. LR: logistic regression; SD: standard deviation; SHAP: SHapley Additive exPlanations.

4 DISCUSSION

4.1 Interpretation of Findings

All four models achieved discrimination above chance, but paired confidence intervals showed no clear XGBoost advantage over LR; equivalence was not established. This aligns with a review finding no evidence that machine-learning methods outperform LR in clinical prediction[17]. Observed-to-expected ratios suggested modest overprediction, but estimates without uncertainty do not establish calibration near the action threshold. Comparisons of death and reintubation used different case groups with common event-free controls; higher discrimination for death therefore does not establish its contribution to composite performance.

4.2 Comparison with Previous Studies

Differences in populations, outcomes, timing and validation limit cross-study comparisons. Unlike our outcome, Zhao et al. included non-invasive ventilation alone as failure[5]; similar discrimination does not imply equivalent clinical performance. In the RSBI-available test subset, our models showed higher observed discrimination than six-hour mean RSBI, comparing multivariable

prediction with a single index without isolating individual predictors' contributions. A meta-analysis recommended combining RSBI with other clinical information[18], but did not evaluate six-hour averages. Our comparison does not establish superiority to bedside assessment incorporating spontaneous breathing trials, airway protection and secretion clearance[2].

4.3 Practical Implications

At the frozen LR threshold, almost half of retained test patients were flagged, mostly as false positives. This prioritised sensitivity but did not establish a clinically optimal threshold. Detected events are not necessarily preventable, and alerts do not equal additional reviews beyond usual care. In the RSBI-available subset, decision-curve analysis suggested potential net benefit over selected comparators at some thresholds, not demonstrated implementation benefit[16]. The model estimates risk among patients undergoing extubation; it has not been validated to determine whether extubation should proceed or which treatment should follow. Such questions require distinguishing prognosis under observed care from outcomes under alternative treatment strategies[19].

4.4 Limitations

Missingness-related selection limits applicability. Excluded patients accounted for 17.9% of the pre-filter test population but 35.0% of its events, so primary estimates cannot represent that entire population. Retained patients could also have missing predictor summaries. Full-cohort training provided no consistent benefit in retained patients, while discrimination remained lower among excluded patients. These analyses neither establish comparable performance across groups nor distinguish information loss from differences in case mix or care processes.

Six-hour means omit trajectories, variability, abnormality timing, and the effect of extreme values were not assessed. XGBoost's emphasis on ICU duration and admission ICU type may reflect case mix and care processes; observation-weighted summaries may reflect measurement practices[20]. Alternative windows were not compared. Extubated patients do not represent all candidates, and predictors may be unavailable at earlier decisions. Single-centre data, incomplete 48-hour observation after early discharge and possible underascertainment of reintubation further limit interpretation.

4.5 Future Work

External validation should assess discrimination, calibration, threshold-specific performance and predictor availability, with reintubation and death also evaluated separately. Earlier use requires confirming predictor availability at the decision point. Prospective studies should assess added value beyond usual assessment and effects on workload, decisions and patient outcomes.

5 CONCLUSION

Six-hour pre-extubation physiological averages and static information provided retrospective prediction of reintubation or in-hospital death within 48 hours in the selected adult ICU cohort. No clear predictive advantage of XGBoost over logistic regression was demonstrated. Missingness-related exclusions limit applicability, and external and prospective evaluation are needed before these findings can support clinical use.

REFERENCES

1. Michelli Marcela Dadam, Aline Braz Pereira, Mariane Ribeiro Cardoso, Tiago Costa Carnin, and Glauco Adrieno Westphal. Effect of reintubation within 48 hours on mortality in critically ill patients after planned extubation. *Respiratory Care*, 69(7): 829–838, 2024. doi: 10.4187/respcare.11077.

2. K. J. Roberts, L. T. Goodfellow, C. M. Battey-Muse, C. A. Hoerr, M. L. Carreon, M. E. Sorg, J. Glogowski, T. D. Girard, N. R. MacIntyre, and D. R. Hess. AARC clinical practice guideline: Spontaneous breathing trials for liberation from adult mechanical ventilation. *Respiratory Care*, 69(7):891–901, 2024. doi: 10.4187/respcare.11735.

3. Vatsal Trivedi, Dipayan Chaudhuri, Rehman Jinah, Joshua Piticaru, Arnav Agarwal, Kuan Liu, Eric McArthur, Michael C. Sklar, Jan O. Friedrich, Bram Rochwerg, and Karen E. A. Burns. The usefulness of the rapid shallow breathing index in predicting successful extubation: A systematic review and meta-analysis. *Chest*, 161(1):97–111, 2022. doi: 10.1016/j.chest.2021.06.030.

4. Tingting Chen, Jun Xu, Haochao Ying, Xiaojun Chen, Ruiwei Feng, Xueling Fang, Honghao Gao, and Jian Wu. Prediction of extubation failure for intensive care unit patients using light gradient boosting machine. *IEEE Access*, 7:150960–150968, 2019. doi: 10.1109/ACCESS.2019.2946980.

5. Qin-Yu Zhao, Huan Wang, Jing-Chao Luo, Ming-Hao Luo, Le-Ping Liu, Shen-Ji Yu, Kai Liu, Yi-Jie Zhang, Peng Sun, Guo-Wei Tu, and Zhe Luo. Development and validation of a machine-learning model for prediction of extubation failure in intensive care units. *Frontiers in Medicine*, 8:676343, 2021. doi: 10.3389/fmed.2021.676343.

6. Péter Bakos, Bence Szabó, Dávid Laczkó, Caner Turan, Shir Galin, Péter Hegyi, László Zubek, András Lovas, and Zsolt Molnár. Machine learning models predicting extubation success in mechanically ventilated patients: A systematic review and meta-analysis. *Intensive Care Medicine Experimental*, 14: 89, 2026. doi: 10.1186/s40635-026-00934-0.

7. Kiyoyasu Fukuchi, Itsuki Osawa, Shunya Satake, Honoka Ito, Junichiro Shibata, Eisuke Dohi, Daisuke Kasugai, Yoshihisa Miyamoto, Hiroyuki Ohbe, Mitsuhiro Tamoto, Naoki Yamada, Keisuke Yoshikawa, and Tadahiro Goto. The contribution of chest X-Ray to predict extubation failure in mechanically ventilated patients using machine learning-based algorithms. *Critical Care Explorations*, 4(6): e0718, 2022. doi: 10.1097/CCE.0000000000000718.

8. Alistair Johnson, Lucas Bulgarelli, Tom Pollard, Brian Gow, Benjamin Moody, Steven Horng, Leo Anthony Celi, and Roger Mark. MIMIC-IV. *PhysioNet*, 2024. doi: 10.13026/kpb9-mt58. URL `https://physionet.org/content/mimiciv/3.1/`. Version 3.1.

9. Alistair E. W. Johnson, Lucas Bulgarelli, Lu Shen, et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10:1, 2023. doi: 10.1038/s41597-022-01899-x.

10. Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '16, pages 785–794, New York, NY, USA, 2016. Association for Computing Machinery. ISBN 9781450342322. doi: 10.1145/2939672.2939785. URL `https://doi.org/10.1145/2939672.2939785`.

11. Sudhir Varma and Richard Simon. Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics*, 7:91, 2006. doi: 10.1186/1471-2105-7-91.

12. John C. Platt. Probabilities for SV Machines. In Alexander J. Smola, Peter Bartlett, Bernhard Schölkopf, and Dale Schuurmans, editors, *Advances in Large-Margin Classifiers*. The MIT Press, September 2000. ISBN 978-0-262-28397-7. doi: 10.7551/mitpress/1113.003.0008. URL `https://doi.org/10.7551/mitpress/1113.003.0008`.

13. Scott M. Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. Explainable AI for trees: From local explanations to global understanding. arXiv preprint arXiv:1905.04610, 2019. URL `https://arxiv.org/abs/1905.04610`.

14. Takaya Saito and Marc Rehmsmeier. The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):1–21, 03 2015. doi: 10.1371/journal.pone.0118432. URL `https://doi.org/10.1371/journal.pone.0118432`.

15. Ben Van Calster, David J. McLernon, Maarten van Smeden, Laure Wynants, and Ewout W. Steyerberg. Calibration: the Achilles heel of predictive analytics. *BMC Medicine*, 17:230, 2019. doi: 10.1186/s12916-019-1466-7.

16. Andrew J. Vickers and Elena B. Elkin. Decision curve analysis: A novel method for evaluating prediction models. *Medical Decision Making*, 26(6): 565–574, 2006. doi: 10.1177/0272989X06295361. URL `https://doi.org/10.1177/0272989X06295361`.

17. Evangelia Christodoulou, Jie Ma, Gary S. Collins, Ewout W. Steyerberg, Jan Y. Verbakel, and Ben Van Calster. A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models. *Journal of Clinical Epidemiology*, 110:12–22, 2019. doi: 10.1016/j.jclinepi.2019.02.004. URL `https://doi.org/10.1016/j.jclinepi.2019.02.004`.

18. Donghui Jia, Hengyang Wang, Qian Wang, Wenrui Li, Xuhong Lan, Hongfang Zhou, and Zhigang Zhang. Rapid shallow breathing index predicting extubation outcomes: A systematic review and meta-analysis. *Intensive and Critical Care Nursing*, 80: 103551, 2024. doi: 10.1016/j.iccn.2023.103551.

19. Nan van Geloven, Sonja A. Swanson, Chava L. Ramspek, Kim Luijken, Merel van Diepen, Tim P. Morris, Rolf H. H. Groenwold, Hans C. van Houwelingen, Hein Putter, and Saskia le Cessie. Prediction meets causal inference: The role of treatment in clinical prediction models. *European Journal of Epidemiology*, 35(7):619–630, 2020. doi: 10.1007/s10654-020-00636-1. URL `https://doi.org/10.1007/s10654-020-00636-1`.

20. Denis Agniel, Isaac S. Kohane, and Griffin M. Weber. Biases in electronic health record data due to processes within the healthcare system: Retrospective observational study. *BMJ*, 361:k1479, 2018. doi: 10.1136/bmj.k1479. URL `https://doi.org/10.1136/bmj.k1479`.

A APPENDIX: COHORT CRITERIA, PREDICTOR DEFINITIONS AND COHORT CHARACTERISTICS

A.1 Cohort Eligibility and Outcome Definitions

Table A.1. Cohort eligibility and outcome operationalisation in MIMIC-IV v3.1.

Criterion or outcome	Source table(s)	Implemented rule
Adult ICU extubation	`derived.icustay_detail`; `icu.procedureevents`	Require `admission_age` ≥ 18 and an extubation procedure whose `starttime` falls between `icu_intime` and `icu_outtime`, inclusive, within the same `stay_id`.
Planned extubation	`icu.procedureevents`; `icu.d_items`	Identify events using item IDs 227194, 225468 and 225477. Retain label `Extubation` (227194); exclude unplanned patient-initiated (225468) and non-patient-initiated (225477) events.
Index extubation	`subject_id`; `extubation_time`; `stay_id`	Retain the earliest planned extubation per patient, ordered by `extubation_time`, then `stay_id`, before applying subsequent eligibility criteria. Define its procedure `starttime` as index time t.
Prior invasive ventilation	`derived.ventilation`	Require a record with the same `stay_id`, `ventilation_status = 'InvasiveVent'` and `starttime` $< t$.
DNR/DNI exclusion	`icu.chartevents`; item ID 223758	Select the latest code-status record in the index ICU stay with `charttime` $\leq t$ and `storetime` $\leq t$, ordered by descending `charttime`, then `storetime`. Exclude values `DNR / DNI`, `DNR` or `DNI`; retain undocumented status.
Physiological data availability	Six-hour physiological summaries	Before imputation, retain patients with $\leq 50\%$ missing values across the 19 physiological summaries in $[t - 6\,\text{h}, t)$, equivalent to at least 10 observed summaries. Static predictors, identifiers and outcomes are excluded from the denominator.
Reintubation within 48 hours	`icu.procedureevents`; item ID 224385	Require a procedure with the same `subject_id` and `hadm_id` and `starttime` in $(t, t + 48\,\text{h}]$. Matching to the index `stay_id` is not required.
In-hospital death within 48 hours	`hosp.admissions`	Require `deathtime` in $(t, t + 48\,\text{h}]$, matched by `subject_id` and `hadm_id`.
Composite extubation failure	Derived outcome	Assign 1 if either recorded reintubation or in-hospital death occurs within the outcome window; otherwise assign 0. Non-invasive ventilation alone does not meet the endpoint.

Notes: Source tables use the `physionet-data.mimiciv_3_1_` prefix. The missingness criterion assesses availability of physiological summaries rather than six hours of continuous measurement. Absence of a recorded outcome does not guarantee complete event-free follow-up after hospital discharge.

A.2 Predictor Definitions and Extraction Rules

Table A.2 lists the 19 physiological summaries used for the patient missingness criterion. Exported physiological column names append `_mean_6h` to the names shown. Missingness is measured before imputation in the retained cohort (n = 17,146). Table A.3 defines the five static candidate predictors. The criteria below describe the implemented extraction filters; they are not clinical reference ranges. Broader plausibility checks in the modelling notebook report values without altering them.

Table A.2. Physiological candidate predictors, extraction rules and retained-cohort missingness.

Feature	Source: item ID(s)	Unit	Accepted values / derivation	Missing (%)
`heart_rate`	CE: 220045	beats/min	$0 < x < 300$	0.05
`mbp`	CE: 220052, 220181, 225312	mmHg	$0 < x < 300$	0.06
`resp_rate`	CE: 220210, 224690	breaths/min	$0 < x < 70$	0.06
`temperature`	CE: 223761, 223762	°C	Fahrenheit: $70 < x < 120$, then $(x - 32)/1.8$; Celsius: $10 < x < 50$. Timestamp means rounded to two decimals.	11.89
`spo2`	CE: 220277	%	$0 < x \leq 100$	0.17
`respiratory_rate_spontaneous`	CE: 224689	breaths/min	$x > 0$	16.15
`tidal_volume_spontaneous`	CE: 224686	mL	$x > 0$	14.20
`tidal_volume_observed`	CE: 224685	mL	$x > 0$	2.71
`minute_volume`	CE: 224687	L/min	$x > 0$	0.59
`peep`	CE: 220339, 224700	cmH_2O	$0 \leq x \leq 100$	0.38
`fio2_chartevents`	CE: 223835	fraction	Retain $0.20 \leq x \leq 1$; divide values $20 \leq x \leq 100$ by 100.	0.52
`ph`	LE: 50820	unitless	$x > 0$; arterial specimens only.	38.67
`pco2`	LE: 50818	mmHg	$x > 0$; arterial specimens only.	38.91
`po2`	LE: 50821	mmHg	$x > 0$; arterial specimens only.	38.92
`bicarbonate`	LE: 50803	mmol/L	$x \geq 0$; arterial specimens only.	99.84
`lactate`	LE: 50813	mmol/L	$0 \leq x \leq 10{,}000$; arterial specimens only.	65.35
`fio2`	LE: 50816	fraction	Retain $0.20 < x \leq 1$; divide values $20 < x \leq 100$ by 100. Arterial specimens only.	89.84
`pao2fio2ratio`	Derived from LE	mmHg	PaO_2/fractional FiO_2 from the same arterial specimen; then average available ratios.	89.86
`RSBI`	Derived from CE	breaths/min/L	Spontaneous respiratory rate divided by spontaneous tidal volume in litres at the same timestamp; then average available ratios.	21.45

Notes: CE: `chartevents`; P/F: PaO_2/FiO_2 ratio; RSBI: rapid shallow breathing index.; LE: `labevents`. All measurements must fall in $[t - 6\,\text{h}, t)$ and have `storetime` $\leq t$. Arterial status requires LE item 52033 with value `ART.`, matched by patient and specimen ID and available by t. Ventilator and blood-gas extraction also require measurement time no earlier than ICU admission. Values failing a feature's rule become missing for that feature. FiO_2 from CE and LE remains separate; no cross-source substitution is used for the P/F ratio. Some implemented filters are broad (e.g., pH and lactate) and should not be interpreted as validation of physiological plausibility.

Table A.3. Static candidate predictors and non-predictive metadata.

Predictor	Unit/type	Definition and source
`admission_age`	years	Admission age from derived `icustay_detail`; cohort restricted to age ≥ 18.
`gender`	categorical	Recorded sex from derived `icustay_detail`, sourced from `patients`.
`icu_type_at_admission`	categorical	`icustays.first_careunit`; describes the first care unit, not necessarily the unit at extubation.
`icu_hours_before_extubation`	hours	Extubation time minus ICU admission time; not total ICU length of stay.
`hospital_hours_before_extubation`	hours	Extubation time minus hospital admission time; not total hospital length of stay.

Notes: `subject_id`, `hadm_id` and `stay_id` are linkage identifiers. `extubation_time`, `anchor_year` and `anchor_year_group` support temporal assignment. `hospital_observed_48h` records whether hospital discharge occurred at least 48 hours after extubation. These fields, outcome labels and observation-count metadata are excluded from model predictors and the 19-feature missingness denominator. Missing feature summaries remain missing in the exports; population-level imputation is learned within training folds.

A.3 Cohort Characteristics

Table A.4. Characteristics of the development and temporal test cohorts, patients excluded by the missingness filter, and the full eligible cohort.

Characteristic	Development	Temporal test	Excluded (all years)	Full eligible (all years)
Patients, n	15,493	897	1,875	19,021
Age, median [IQR]	66 [55–76]	65 [54–74]	64 [51–78]	66 [55–76]
Female, n (%)	5,899 (38.1)	319 (35.6)	827 (44.1)	7,364 (38.7)
ICU hours before extubation, median [IQR]	22.3 [10.6–65.6]	40.6 [15.8–101.6]	33.8 [14.6–94.3]	23.9 [11.2–70.6]
Hospital hours before extubation, median [IQR]	49.1 [22.4–126.0]	69.8 [34.0–164.5]	46.6 [18.4–140.1]	50.6 [22.6–131.9]
ICU type at admission, n (%)				
Cardiac vascular ICU	6,144 (39.7)	328 (36.6)	69 (3.7)	6,746 (35.5)
Medical ICU	2,404 (15.5)	124 (13.8)	513 (27.4)	3,200 (16.8)
Surgical ICU	2,202 (14.2)	76 (8.5)	324 (17.3)	2,701 (14.2)
Trauma surgical ICU	2,025 (13.1)	113 (12.6)	341 (18.2)	2,546 (13.4)
Medical/surgical ICU	1,424 (9.2)	63 (7.0)	274 (14.6)	1,854 (9.7)
Coronary care unit	872 (5.6)	57 (6.4)	173 (9.2)	1,152 (6.1)
Neurosurgical ICU	195 (1.3)	54 (6.0)	105 (5.6)	384 (2.0)
Neuro intermediate	124 (0.8)	55 (6.1)	40 (2.1)	255 (1.3)
Neuro stepdown	100 (0.6)	0 (0.0)	12 (0.6)	116 (0.6)
Surgery/vascular/intermediate	0 (0.0)	14 (1.6)	17 (0.9)	37 (0.2)
Other	3 (0.0)	13 (1.4)	7 (0.4)	30 (0.2)
Extubation failure, n (%)	1,503 (9.7)	106 (11.8)	653 (34.8)	2,348 (12.3)
Death within 48 hours, n (%)	720 (4.6)	58 (6.5)	580 (30.9)	1,410 (7.4)
Reintubation within 48 hours, n (%)	801 (5.2)	51 (5.7)	76 (4.1)	964 (5.1)
Negative labels without complete 48-hour observation, n	496	23	206	742
Model predictors missing, mean (%)	21.9	24.0	45.2	24.3

Notes: Values are counts (percentages) unless otherwise stated. Excluded patients failed the physiological missingness criterion. The full eligible cohort includes retained and excluded patients across all years; 756 retained patients with calendar-year intervals spanning 2020 were omitted from development and temporal testing. Death and reintubation can overlap. Predictor missingness is measured before imputation across model predictors, whereas eligibility uses only the 19 physiological summaries. IQR: interquartile range.

B MODEL HYPERPARAMETERS AND CLASS WEIGHTS

Table B.1. Model hyperparameters and class weights.

Model	Hyperparameters	Class weights
Tuned LR	$C \in \{0.001, 0.01, \mathbf{0.1}^{\dagger}, 1, 10, 100\}$	Balanced: $n/(2n_k)$
Untuned LR	$C = 1$	Balanced: $n/(2n_k)$
Tuned XGBoost	Trees $\in \{200, \mathbf{400}^{\dagger}\}$ Maximum depth $\in \{2, \mathbf{3}, 4^{\dagger}\}$ Learning rate $\in \{\mathbf{0.03}^{\dagger}, 0.1\}$ Minimum child weight $\in \{1, \mathbf{5}^{\dagger}\}$ L2 penalty $\lambda \in \{1, \mathbf{10}^{\dagger}\}$	Positive: n_-/n_+ Negative: 1
Untuned XGBoost	Trees = 250 Maximum depth = 3 Learning rate = 0.05 Minimum child weight = 1 L2 penalty $\lambda = 1$	Positive: n_-/n_+ Negative: 1

Notes: Weights are calculated within each training set. n is the total sample size; n_k is the number of patients in class k; n_- and n_+ are the negative and positive outcome counts.

Bold and † indicate selected values for the primary filtered-cohort and full-cohort models, respectively. Both XGBoost variants used row and column subsampling of 0.8. LR: logistic regression.

GENERATIVE AI DECLARATION

Use of generative AI: GenAI is used in this report.

Tool(s): Claude (claude.ai) and ChatGPT (openai.com)

Purpose and scope: Both GenAI are used for brainstorming, planning, checking sentence structure and grammar in both the report and code.

Declaration: GenAI is not used to create text or code.

AUTHOR CONTRIBUTIONS

Table B.2. Author contributions.

Author	Contributions
Jiajian He	Data curation (data preprocessing); Methodology (cohort phenotyping).
Wenhao Duan	Writing – original draft (Abstract and Introduction); Writing – review and editing.
Haoran Zhang	Writing – original draft (Discussion); Writing – review and editing.
Kun Xing	Methodology; Software (model development and implementation).
Zongcheng Jiang	Formal analysis (analysis of model results).